Event Driven Architecture with SAP Event Mesh

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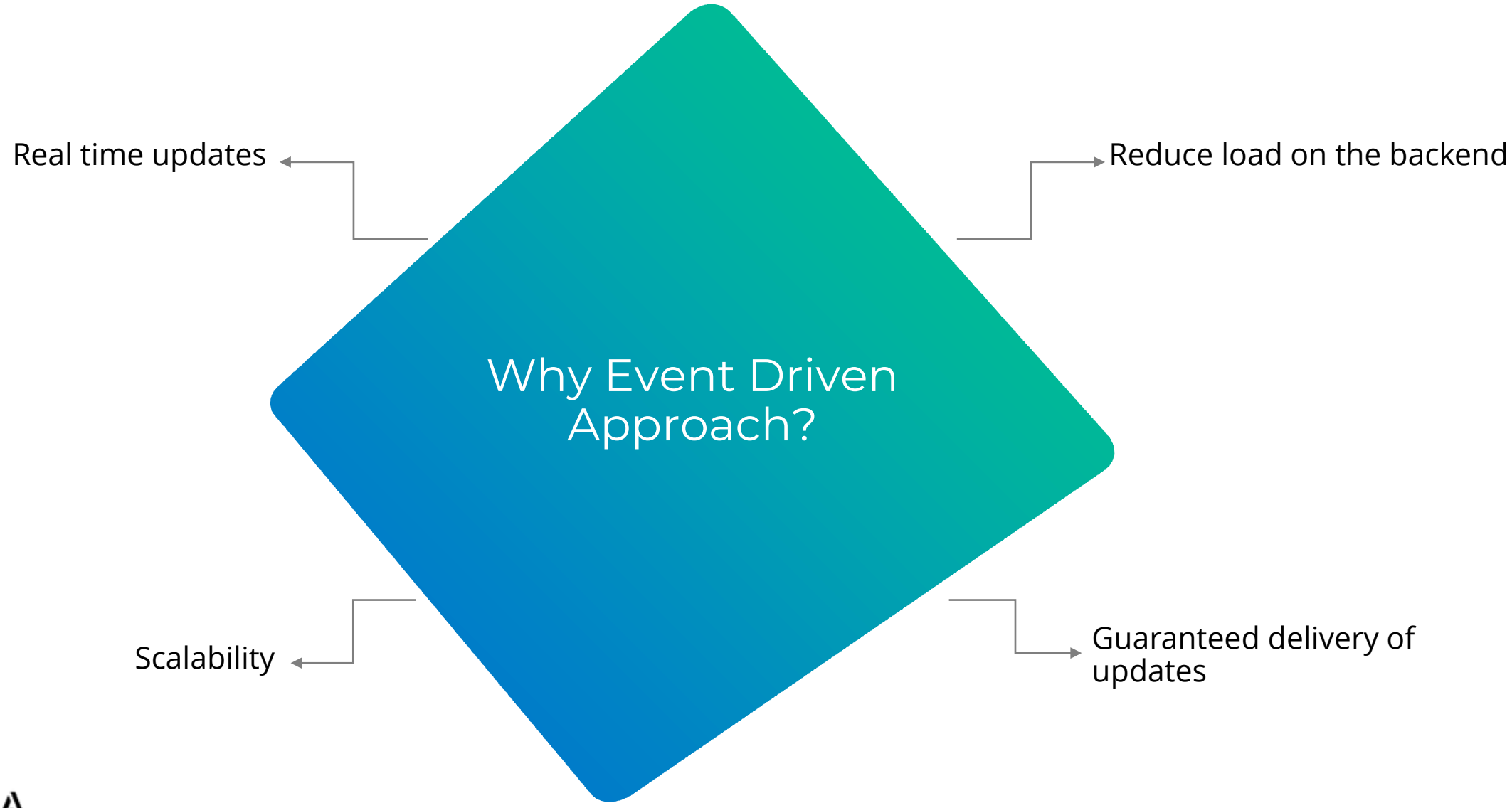
SANDESH KURUMELLA

We're going to talk about:

- Why an Event Driven Approach?
- What is it about?
- How does SAP Event Mesh address this approach?
- Takeaways
- Q&A

How it started?



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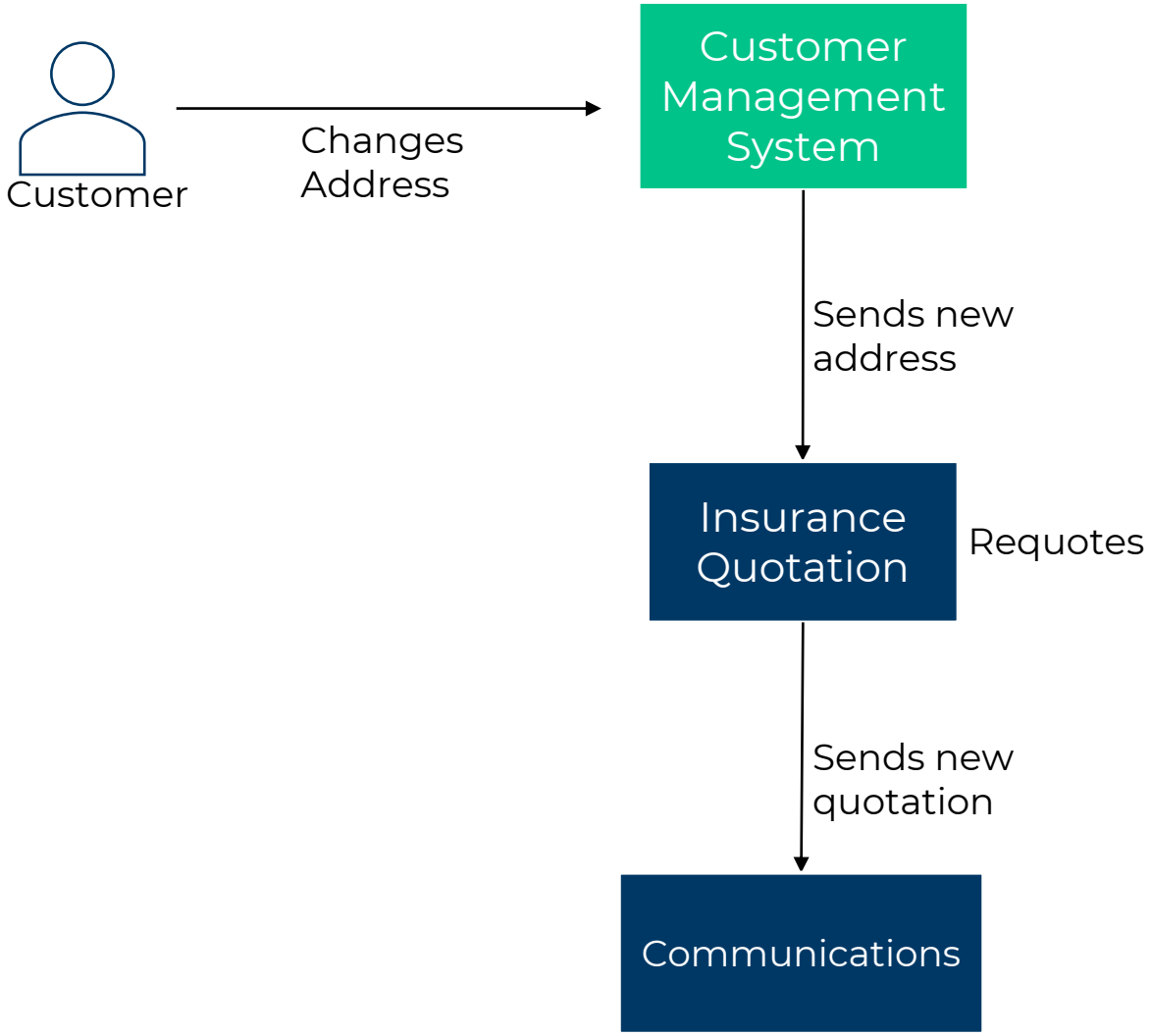
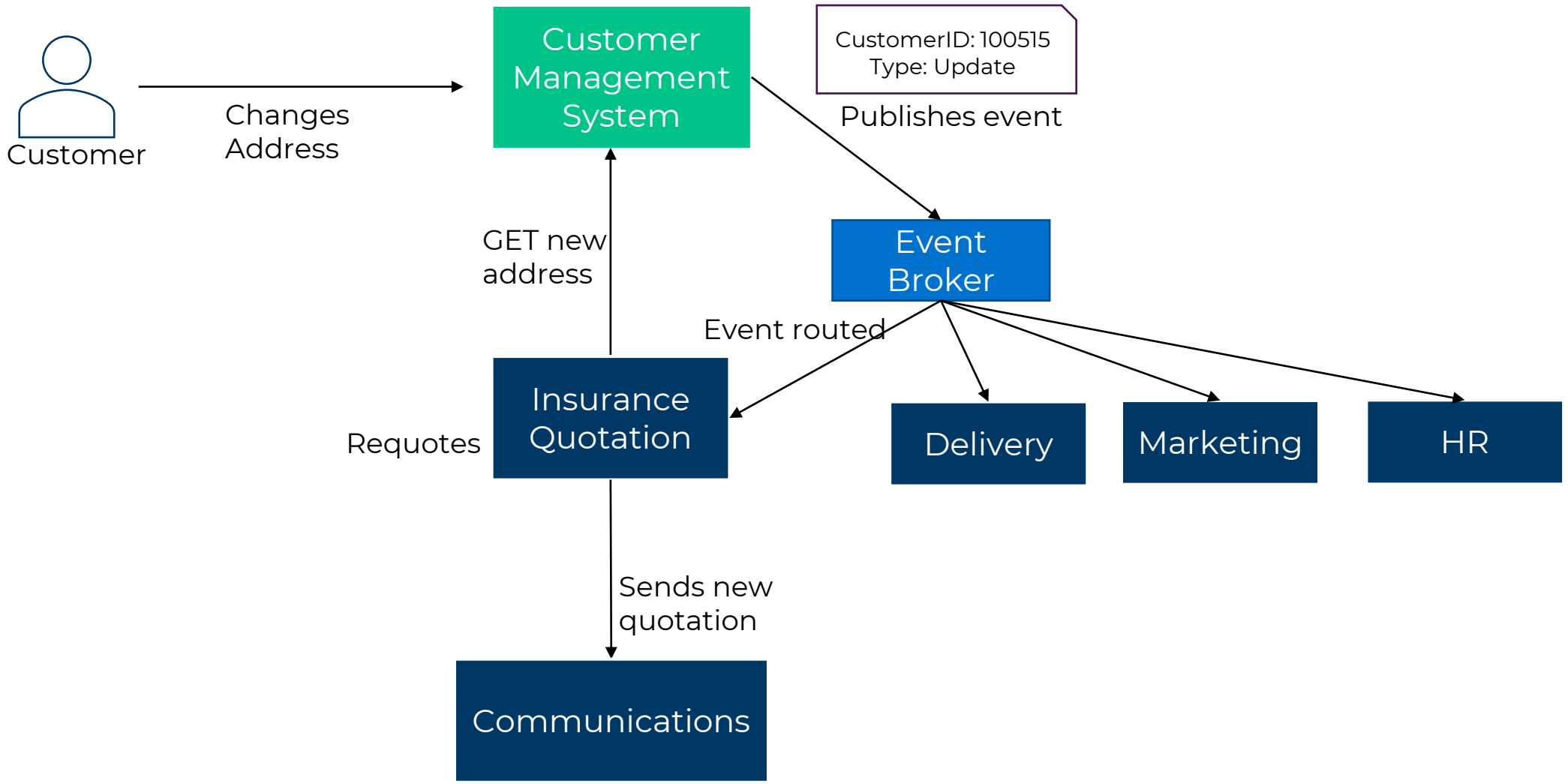
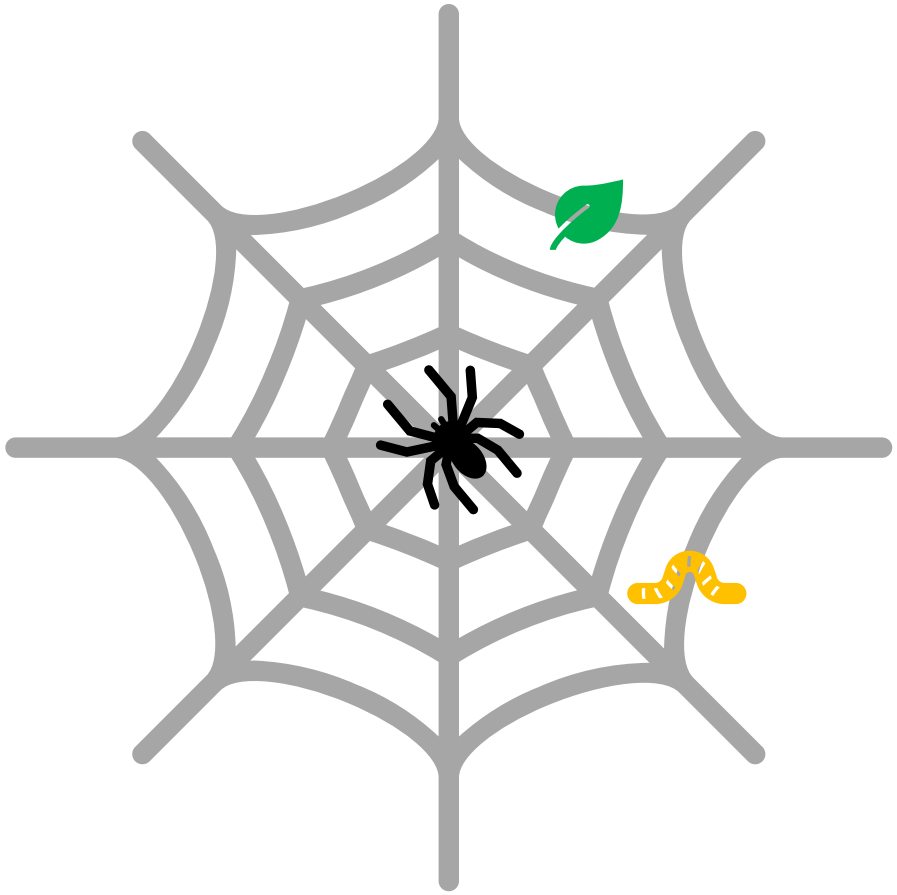
Traditional Architecture



Event Driven Architecture



Event-Driven Architecture



- An **event** is a significant change in state
- Notification Events or Data Events can be sent from the **event source** to inform about the change
- A message holding the event description is sent via an **event broker**
- **Event consumers** have registered with the event broker and they are informed of the event

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SAP’s event-driven ecosystem

Event sources

SAP S/4HANA
SAP S/4HANA Cloud
SAP ERP
SAP SuccessFactors
SAP Marketing Cloud
Other SAP solutions
3rd party

Eventing infrastructure

SAP Event Mesh

- Event broker
- Powered by Solace PubSub+
- Event routing
- Event catalogs

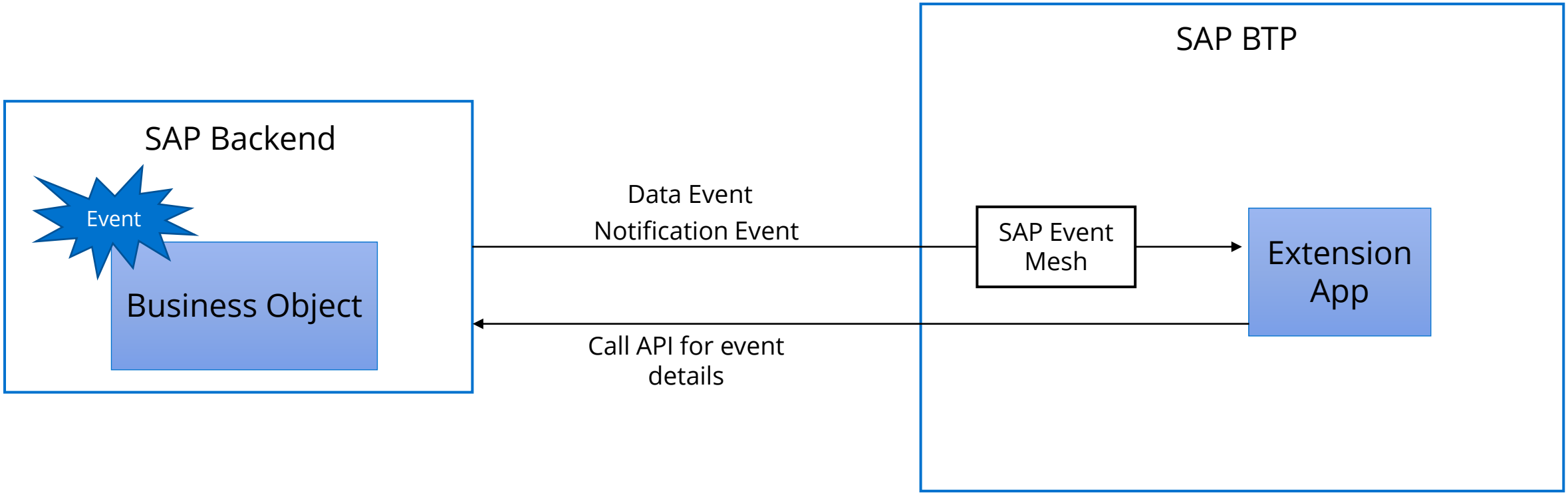
Event consumer

SAP Integration Suite
SAP Extension Suite
SAP BTP Services
SAP back ends
Applications
3rd party

Notification event vs Data Event

Notification event contains minimal data set required to inform of a significant change

Data event contains full data set of business object for which significant change has happened



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Event payload

- Technically an event can contain any format of data
- Events format in SAP is based on Cloud Event specification (cloudevents.io). The same format is used by other vendors like Microsoft, Google etc.


```
{
"specversion" : "1.0",
"type" : “sap.s4.beh.businesspartner.v1.BusinessPartner.Changed.v1",
"source“: ”/default/Amista.s4.demo/01",
"id" : “fa384n-e29fnn-ks8111n-hfejj139-fj99hk",
“time” : "2022-01-22T17:50:44.428Z",
“data” :
{“objectType”: “BUPA”,
“event”: “CHANGED”,
“objectId”: “0450201836”}
}
```


Notification event


```
{
"specversion" : "1.0",
"type" : “sap.s4.beh.businesspartner.v1.BusinessPartner.Changed.v1",
"source“: ”/default/Amista.s4.demo/01",
"id" : “fa384n-e29fnn-ks8111n-hfejj139-fj99hk",
“time” : "2022-01-22T17:50:44.428Z",
“data” :
{“objectType”: “BUPA”,
“event”: “CHANGED”,
“objectId”: “0450201836”,
“salesOrganisation”:”1010”,
“distributionChannel”:”00”,
“division”:”02”}
}
```


Data event/Decision event

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Notification vs Data Events

Notification Event

Advantages

- Very **small**
- Controlled data access

Challenges

- **Suitable API required** on top of the event
- Additional configuration for API access

Data Event

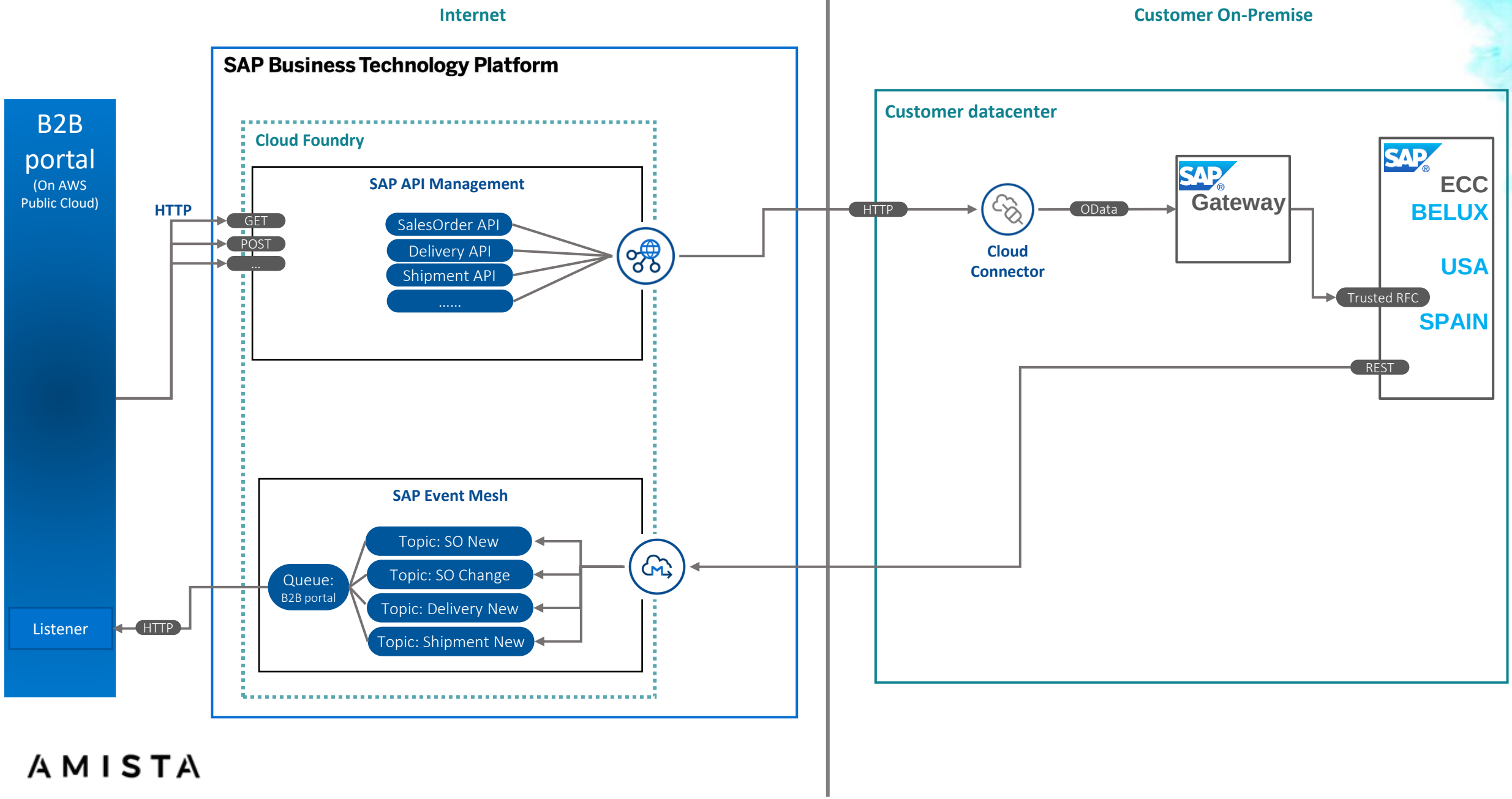
Advantages

- All required **data included** in event
- Different flavors
  - Full Data Event
  - Decision Event

Challenges

- **Bigger** in size
- Data access and protection
- **High resource consumption** for source, event broker and consumer

Architecture



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Approach on SAP ECC

1

Install **SAP Event Enablement** Add-on

2

Configure **SWE2** (BOR events)

Write custom ABAP in BOR extension

3

Configure Event Enablement Add-on

4

Link BO & Event enablement add-on by message type

5

Configure **SAP Event Mesh**

Takeaways

- Supports exchange patterns of **Pub/Sub** and **Point to Point**
- **Publish business events** from SAP and non-SAP sources across hybrid landscapes
- Supports communication protocols like AMQP, MQTT and HTTPS
  - AMQP – Within Cloud Foundry
  - MQTT – With on-premise systems
  - Java and NodeJS libraries available for AMQP and MQTT
- Real time replication is possible
- Part of the **SAP Integration Suite and SAP Extension Suite** to implement extension and integration scenarios
- Monitoring of the events is not great

New offering

SAP ADVANCED EVENT MESH

Q & A

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